

Detection and Correction of Silent Decision Degradation in AI-Assisted Logistics Systems

Abstract

AI-assisted logistics and supply chain systems are increasingly deployed in complex operational environments such as defense logistics, transportation planning, and industrial production systems. Current evaluation approaches primarily focus on static performance metrics such as accuracy, cost efficiency, or route optimality. However, in real-world operations characterized by partial observability, delayed information, evolving constraints, and shifting priorities, AI systems may continue to produce plausible outputs while their decision quality degrades over time. This phenomenon, referred to as **silent decision degradation**, remains poorly understood and inadequately addressed by existing evaluation practices.

This project conducts a systematic, simulation-driven analysis of silent decision degradation in AI-assisted logistics systems. Controlled operational disturbances are introduced to study how degradation emerges and propagates under realistic conditions. The analysis demonstrates the limitations of conventional metrics in identifying gradual decision failure and explores alternative behavioral and structural indicators such as decision volatility, constraint violation trends, and confidence–outcome divergence. Based on these analytical findings, the project derives a **decision integrity assessment framework** and implements a prototype analytical tool that translates degradation patterns into actionable system-level indicators. The final outcome bridges analysis and system design by converting empirical insights into a reusable evaluation framework for improving long-term reliability in AI-assisted logistics systems.

Problem Statement

AI-driven logistics and supply chain decision systems are typically designed and evaluated using static performance metrics under idealized assumptions of stable data distributions and complete information. In practical production environments, these assumptions rarely hold. Systems operate under incomplete data, delayed updates, dynamic constraints, and evolving operational priorities. Under such conditions, AI models may not fail explicitly; instead, they experience **silent decision degradation**, where decision quality deteriorates gradually without triggering conventional performance alarms.

Existing research lacks structured analytical methods to systematically study this form of degradation and to translate analytical insights into practical system-level evaluation mechanisms. As a result, decision reliability risks remain hidden until significant operational impact occurs.

The problem addressed in this project is twofold:

1. the absence of effective analytical frameworks to detect and characterize silent decision degradation in AI-assisted logistics systems, and
2. the lack of a structured mechanism to convert analytical findings into a tangible, reusable decision integrity assessment tool.

The objective is to analyze degradation behavior under realistic disturbances and, based on this analysis, develop a prototype framework that operationalizes degradation indicators into a product-oriented outcome for evaluating decision reliability in AI-assisted logistics systems.

Objectives of the Project

The primary objective of this project is to **analyze the phenomenon of silent decision degradation** in AI-assisted logistics systems and translate analytical insights into a practical decision integrity assessment framework.

The specific objectives are:

1. To design a realistic simulation environment that models AI-assisted logistics decision-making under dynamic and uncertain operating conditions.
2. To systematically introduce controlled operational disturbances such as partial observability, delayed data, and shifting priorities to study their impact on decision quality over time.

3. To analyze the limitations of conventional performance metrics in detecting gradual degradation in AI-driven decision systems.
 4. To identify and quantify alternative behavioral and structural indicators that better characterize decision degradation.
 5. To synthesize analytical findings into a structured framework and prototype tool for assessing decision integrity in AI-assisted logistics systems.
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Methodology

The project follows a structured, analysis-driven methodology consisting of four main phases:

1. System Modeling and Simulation

A representative AI-assisted logistics decision system is modeled using a combination of machine learning, rule-based logic, and optimization techniques. The system simulates core logistics functions such as routing, prioritization, or allocation, ensuring relevance to real-world production and defense-oriented scenarios.

2. Operational Disturbance Injection

Controlled disturbances are introduced into the simulation to emulate realistic operational challenges, including incomplete data availability, delayed updates, distribution shifts, and dynamic constraint changes. These disturbances are designed to induce gradual, non-catastrophic degradation in decision quality.

3. Analytical Evaluation of Degradation

System behavior is analyzed over time using both conventional performance metrics and alternative indicators such as decision volatility, constraint violation frequency, structural changes in decision graphs, and confidence–outcome divergence. Comparative analysis is conducted to evaluate the effectiveness of each indicator in detecting degradation.

4. Framework Derivation and Prototype Development

Insights from the analytical phase are consolidated into a decision integrity assessment framework. A prototype analytical tool is developed to

operationalize the framework, enabling systematic evaluation and visualization of decision degradation patterns in AI-assisted logistics systems.

Expected Outcomes

The expected outcomes of the project include:

1. A validated simulation framework demonstrating how silent decision degradation emerges in AI-assisted logistics systems under realistic operational disturbances.
 2. Empirical evidence highlighting the inadequacy of conventional performance metrics in detecting gradual decision degradation.
 3. Identification of robust behavioral and structural indicators that reliably signal degradation before significant performance collapse.
 4. A structured decision integrity assessment framework derived from analytical findings.
 5. A prototype analytical artifact (framework or tool) that translates analysis into a product-oriented outcome for evaluating decision reliability.
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System Definition — Service-Level-Constrained Inventory Decision under Drift

1. What decision does the system make?

At each decision epoch t (weekly), the system determines the **inventory replenishment quantity** for each product at each store, such that:

- expected total inventory cost is minimized, and
- a predefined **service level constraint** is satisfied,
- under **uncertain and time-varying demand and lead-time conditions**.

Formally (conceptual, not mathematical):

Decision:

Select reorder quantity Q_t conditioned on the current system state, where system parameters are non-stationary and partially observable.

This decision is **not** a closed-form calculation because:

- demand distribution parameters drift over time,
- lead-time distributions change in mean and variance,
- service-level feasibility itself evolves,
- cost trade-offs are asymmetric and time-dependent.

2. What inputs does the decision see?

The decision is based on a **state representation** composed of four classes of inputs.

A. Demand State

- Historical demand over recent time windows
- Demand forecast for the next horizon
- Forecast uncertainty indicators (variance, bias trends)

B. Inventory State

- Current on-hand inventory
- Inventory position (on-hand + pipeline – backorders)
- Recent stockout occurrences

C. Lead-Time State

- Estimated lead-time distribution parameters
- Observed lead-time drift indicators
- Variability trends in lead-time realization

D. Cost & Constraint State

- Holding cost

- Shortage or penalty cost
- Target service level (fill rate or cycle service level)

These inputs are **observed imperfectly** and **evolve over time**, making static optimization invalid.

3. What does a “good decision” mean?

A decision is considered **good** if it satisfies all three criteria **simultaneously**:

1. Service-Level Satisfaction

The achieved service level meets or exceeds the target threshold over time, not just instantaneously.

2. Cost Efficiency

Total expected cost (holding + shortage + ordering impact) is minimized without aggressive overstocking.

3. Stability under Drift

Decision quality remains robust as demand and lead-time characteristics gradually change, without requiring frequent retuning.

Crucially, a good decision **does not necessarily minimize cost at every step**—it maintains *decision integrity* under evolving uncertainty.

4. What does the system output?

The system outputs:

- **Reorder quantity Q_{tQ_tQt}** for each product–store pair at decision epoch t .

Optionally (for analysis and diagnostics, not control):

- Predicted service-level risk
- Confidence score associated with the decision

Only Q_{tQ_tQt} directly affects the operational system; all other outputs support analysis and degradation detection.

Why Inventory Replenishment Systems?

Inventory replenishment systems represent a class of decision-making problems that are mathematically well studied yet operationally fragile. Classical inventory models assume stable demand distributions, known lead times, and static cost parameters, conditions that rarely hold in real production environments. In practice, inventory decisions are made under partial observability, delayed feedback, and gradually evolving uncertainty. Importantly, inventory systems typically do not fail abruptly; instead, they absorb suboptimal decisions over time through buffering and slack, causing performance deterioration to manifest slowly and often invisibly. This makes inventory replenishment an ideal domain for studying **silent decision degradation**, where decision quality erodes without triggering obvious performance alarms. By focusing on inventory systems, this work deliberately examines a canonical and widely deployed decision framework to demonstrate that even well-understood and seemingly stable operational systems can experience undetected degradation when exposed to realistic demand and lead-time drift. This choice emphasizes decision integrity over algorithmic novelty and allows analytical insights to generalize across a broad range of production, logistics, and supply chain applications.

While classical inventory models provide valuable analytical structure, their practical deployment requires continuous estimation of uncertain and time-varying system states such as demand distributions, lead-time variability, and service-level feasibility. In this context, machine learning is not used to replace established decision logic, but to **infer evolving system characteristics** that cannot be reliably captured through static assumptions or fixed parameter models.

Why Machine Learning Is Necessary in This Study

Machine learning is employed in this study to address the limitations of classical inventory models under non-stationary and partially observable operating conditions. In real inventory systems, key decision parameters such as demand patterns, forecast uncertainty, and lead-time distributions evolve gradually over time and are only observable through noisy realizations. Static parameter estimation or closed-form solutions assume stationarity and known distributions,

assumptions that degrade silently as operating conditions change. Machine learning provides a flexible mechanism to estimate these evolving state variables from historical data, enabling the decision system to operate under realistic uncertainty without explicitly encoding brittle analytical assumptions. Importantly, ML is used to support state estimation rather than to directly optimize decisions, ensuring that observed degradation arises from environmental drift rather than model instability.

Phase 0: Environment Characterization and Problem Scoping

1. Purpose of Phase 0

The purpose of Phase 0 was to **understand and characterize the operational environment** in which inventory decisions are made, prior to defining or implementing any decision logic.

This phase deliberately avoids optimization, prediction, or control in order to prevent premature assumptions about decision structure or system behavior.

Phase 0 serves as the foundation for all subsequent phases by identifying structural properties, behavioral heterogeneity, and latent risks that influence decision quality but are often overlooked by conventional performance metrics.

2. Necessity of Phase 0

Inventory decision systems are frequently evaluated using aggregate metrics such as forecast accuracy, cost, or service level. However, in real-world supply chain environments:

- Entities are strongly interdependent
- Demand patterns vary across locations and time
- Behavioral responses (e.g., to holidays) are heterogeneous
- Failures may propagate silently through coupled structures

Without explicitly characterizing these properties, any downstream decision system risks optimizing the wrong objective or masking early degradation.

Phase 0 was therefore necessary to:

- Expose structural coupling across stores and departments
 - Identify heterogeneity in demand and behavioral responses
 - Determine whether static statistics adequately represent system dynamics
 - Establish realistic expectations for decision-system behavior
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3. Problem Statement (Phase 0 Perspective)

Given a large-scale retail inventory dataset, the problem addressed in Phase 0 is:

To determine whether the inventory decision environment exhibits structural and behavioral properties that can give rise to silent degradation in decision quality, even when conventional performance metrics remain stable.

This formulation shifts focus from prediction accuracy to **decision integrity under complexity**.

4. Methodology: How Phase 0 Was Conducted

Phase 0 employed exploratory and structural analysis techniques without introducing any decision logic. The methodology consisted of:

4.1 Structural Network Analysis

- Construction of bipartite Store–Department networks to examine demand coupling
- Store–Store similarity networks based on demand composition
- Department-level volatility similarity networks to assess inter-departmental dependence

4.2 Behavioral Analysis

- Measurement of store-specific holiday demand amplification

- Visualization of behavioral similarity through sparse similarity networks

4.3 Statistical Characterization

- Analysis of demand concentration across departments
- Degree and dependency statistics to quantify systemic exposure

All analyses were conducted using aggregated, historical data and were strictly descriptive.

5. Key Observations from Phase 0

5.1 Strong Structural Coupling

Most stores share a common subset of high-volume departments. A small number of departments are connected to nearly all stores, creating dense coupling within the system and increasing the potential for cascading effects.

5.2 Heterogeneous Store Demand Regimes

Store–store similarity analysis reveals distinct clusters, indicating multiple latent demand regimes. This undermines the assumption that a single global decision policy can serve all stores effectively.

5.3 Weak Inter-Departmental Volatility Structure

Department-level volatility networks remain sparse and fragmented, even after smoothing. This negative result demonstrates that volatility alone does not encode meaningful decision dependencies.

5.4 Store-Specific Behavioral Responses

Holiday demand amplification varies significantly across stores, forming sparse behavioral similarity networks. Holiday effects act as localized behavioral fingerprints rather than uniform seasonal multipliers.

5.5 Concentration Risk

A substantial fraction of store demand is concentrated within a small subset of departments, amplifying systemic risk when environmental conditions shift.

6. Interpretation of Phase 0 Findings

Phase 0 establishes that the inventory environment is:

- **Densely coupled**, enabling error propagation
- **Heterogeneous**, with multiple latent operating regimes
- **Behaviorally diverse**, particularly in response to holidays
- **Structurally fragile**, due to demand concentration

These properties create conditions under which decision quality can degrade gradually and silently, without immediate collapse in aggregate metrics.

7. Phase 0 Conclusion

Phase 0 successfully characterizes the decision environment and demonstrates that static performance indicators and isolated statistical measures are insufficient to capture the system's underlying complexity.

The findings motivate the need for:

- Explicit decision modeling (Phase 1)
- Controlled stress and failure injection (Phase 2)
- Behavioral and structural degradation detection (Phase 3)

With the environment fully characterized and scoped, the project transitions from **environment understanding** to **decision-system analysis**.

Phase 0 is complete and frozen.

Phase 1: Problem Formulation — Baseline Inventory Decision System

1. Objective of Phase 1

The objective of Phase 1 is to implement a **simple, executable baseline inventory decision system** that operates within the complex, heterogeneous environment identified in Phase 0. This system is not intended to be optimal. Its sole purpose is

to produce **consistent, repeatable decisions over time** that can later be stress-tested to reveal silent decision degradation.

2. Decision Definition

At each decision epoch (weekly), the system determines:

The quantity of inventory to allocate to each store–department pair for the upcoming week.

This decision is made independently at each time step but under shared system-level constraints.

3. Decision Scope and Granularity

- **Temporal resolution:** Weekly
- **Spatial resolution:** Store × Department
- **Decision horizon:** One-step ahead (no multi-week planning)
- **Control variables:** Inventory allocation quantities

The system does not model ordering lead times, pricing, or promotions at this stage.

4. Input Variables

The decision system observes the following inputs at time (t):

Demand History

- Historical weekly sales for each store–department pair
- Rolling demand statistics (mean, variance)

Temporal & Calendar Signals

- Holiday indicator for the upcoming week
- Seasonal position inferred from historical patterns

Store Characteristics

- Store type
- Store size (proxy for capacity and demand scale)

System Constraints

- Total available inventory capacity (global or per department)
- Non-negativity and feasibility constraints on allocations

Importantly, **no future demand is assumed to be known**.

5. Output Variables

The system outputs:

- $(Q_{s,d,t})$: Allocated inventory quantity for store (s), department (d), at week (t)

These outputs are logged for each decision epoch to enable longitudinal analysis.

6. Baseline Decision Logic

The baseline system uses a **rule-based demand projection** combined with proportional allocation:

1. Estimate expected demand using recent historical averages:

$$\begin{aligned} &[\\ &\hat{D}_{s,d,t} = \text{RollingMean}_k(\text{WeeklySales}_{s,d}) \\ &] \end{aligned}$$

2. Apply a simple seasonal or holiday adjustment factor:

$$\begin{aligned} &[\\ &\hat{D}_{s,d,t}^{adj} = \begin{cases} \hat{D}_{s,d,t} \times \alpha_s & \text{if holiday} \\ \hat{D}_{s,d,t} & \text{otherwise} \end{cases} \\ &] \end{aligned}$$

3. Allocate inventory proportionally under capacity constraints:

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$$Q_{\{s,d,t\}} = \min\left(\hat{D}_{\{s,d,t\}}^{\text{adj}}, \text{AvailableCapacity}_d\right)$$

This logic intentionally avoids optimization or learning to ensure interpretability and stability.

7. Definition of a “Good” Decision

A decision is considered good if it satisfies the following criteria:

- Meets a substantial fraction of realized demand
- Respects inventory and feasibility constraints
- Produces stable allocations over time
- Avoids extreme fluctuations without strong justification

No single scalar objective (e.g., cost minimization) is optimized in Phase 1.

8. Evaluation Metrics (Baseline Only)

The following metrics are tracked but **not optimized**:

- Service level (fulfilled demand / realized demand)
- Allocation stability (week-to-week change magnitude)
- Constraint violation frequency
- Demand mismatch trends

These metrics establish a reference against which degradation can later be detected.

9. Role of Phase 1 in the Overall Workflow

Phase 1 provides:

- A fully operational decision system
- Logged decision trajectories
- A stable baseline suitable for controlled failure injection

The system is deliberately simplistic to ensure that any observed degradation in later phases arises from environmental stress and system behavior rather than model complexity.

Phase 1: Baseline Inventory Decision System

1. Purpose of Phase 1

The objective of Phase 1 was to construct a **simple, transparent, and executable baseline inventory decision system** that operates within the complex environment characterized in Phase 0.

This phase deliberately avoids optimization, machine learning, or advanced forecasting in order to establish a **neutral reference system** whose behavior can later be analyzed under stress.

Rather than seeking optimal performance, Phase 1 focuses on **decision consistency, interpretability, and traceability**, ensuring that any degradation observed in later phases can be attributed to environmental or structural factors rather than model complexity.

2. Why a Baseline Decision System Was Necessary

Phase 0 revealed that the inventory environment exhibits:

- Strong coupling across stores and departments
- Heterogeneous demand regimes
- Concentration risk in a small subset of departments
- Store-specific behavioral responses to holidays

In such an environment, directly introducing complex models risks conflating **model error** with **decision degradation**.

Therefore, Phase 1 establishes a baseline system with **explicit assumptions**, minimal intelligence, and deterministic behavior, serving as a controlled substrate

for subsequent failure analysis.

3. Problem Formulation Recap

At each weekly decision epoch, the system determines:

The quantity of inventory to allocate to each Store–Department pair for the upcoming week.

Inputs

- Historical weekly sales
- Store metadata (type, size)
- Calendar signal (holiday indicator)
- Aggregated exogenous features (temperature, CPI, fuel price, unemployment, markdowns)

Outputs

- Allocated inventory quantity per Store–Department–Week

The decision horizon is **single-step (one week ahead)**, with no future demand visibility and no multi-period optimization.

4. Baseline Decision Logic

The baseline system follows a **rule-based demand projection and allocation strategy**:

1. Demand Estimation

Expected demand is estimated using a rolling mean of historical weekly sales for each Store–Department pair.

If insufficient history exists, a global department-level mean is used as a fallback.

2. Holiday Adjustment

A store-specific holiday amplification factor, derived from historical holiday vs non-holiday sales ratios, is applied during holiday weeks.

3. Capacity Constraint

Allocated inventory is capped using a simple capacity assumption proportional to historical demand levels, ensuring feasibility.

4. Allocation Rule

Inventory is allocated as the minimum of expected demand and capacity, producing deterministic and reproducible decisions.

This logic prioritizes **interpretability and stability** over optimality.

5. Data Integrity and Resolution Alignment

To ensure correctness at the decision level, exogenous features were aggregated to a **unique Store–Date resolution** prior to merging with sales data.

This prevents artificial duplication of decisions and ensures that each Store–Department–Week corresponds to a single, well-defined decision state.

After aggregation and merging:

- No duplicate Store–Department–Date entries were present
- The final decision dataset contained 421,570 unique weekly decision records

This alignment is critical for longitudinal decision analysis.

6. Evaluation Metrics (Observed, Not Optimized)

The following metrics were tracked to characterize baseline behavior:

- **Average service level:** ~0.92
- **Total unmet demand:** $\sim 5.16 \times 10^8$
- **Total overstock:** $\sim 5.64 \times 10^8$
- **Mean weekly allocation:** ~16,100 units
- **Allocation volatility:** ~21,900 units

Service level was defined only for periods with positive realized demand, preserving metric validity.

No metric was optimized or tuned in Phase 1.

7. Key Observations from Phase 1

7.1 Acceptable Average Performance with High Inefficiency

The system achieves a reasonable average service level while simultaneously exhibiting large unmet demand and overstock volumes.

This indicates that acceptable aggregate performance can coexist with significant inefficiency.

7.2 Decision Volatility Exceeds Mean Allocation

Allocation volatility exceeds the mean allocation magnitude, revealing that decisions fluctuate sharply week-to-week despite stable input logic.

This suggests **latent fragility**, where small changes in inputs can induce disproportionate decision shifts.

7.3 Presence of Extreme Local Behaviors

Extreme service-level values occur in low-demand regimes and are retained as indicators of instability rather than treated as data errors.

These regimes represent early stress points within the system.

8. Interpretation and Significance

Phase 1 demonstrates that even a simple, transparent decision system operating under realistic constraints can exhibit:

- High decision volatility
- Inefficiency masked by acceptable average metrics
- Sensitivity to localized demand conditions

These characteristics create ideal conditions for **silent decision degradation**, where decision quality deteriorates gradually without triggering immediate metric

collapse.

9. Phase 1 Conclusion

Phase 1 successfully establishes a deterministic baseline inventory decision system that is:

- Executable and auditable
- Interpretable and assumption-driven
- Stable in logic but fragile in behavior

This baseline provides a controlled foundation for systematically introducing stress and observing how decision integrity degrades over time.

Phase 1 is complete and frozen.

Phase 2: Failure Injection and Silent Decision Degradation Analysis

1. Purpose of Phase 2

The purpose of Phase 2 is to **systematically expose and characterize silent decision degradation** in the baseline inventory decision system developed in Phase 1. Unlike overt system failures, silent degradation refers to gradual deterioration in decision quality that occurs without immediate or obvious collapse in conventional performance metrics.

This phase shifts focus from *decision correctness* to **decision behavior under stress**, investigating how stable decision logic can produce increasingly suboptimal or unstable outcomes when its underlying assumptions are violated.

2. Necessity of Phase 2

Phase 1 demonstrated that the baseline system achieves acceptable average performance while exhibiting high allocation volatility and inefficiency. However,

such aggregate metrics are insufficient to determine whether decision integrity is being preserved over time.

In real-world inventory systems:

- Input signals are delayed, incomplete, or biased
- Demand distributions evolve gradually
- Behavioral patterns shift asymmetrically across entities
- Structural coupling amplifies local disturbances

These conditions do not typically trigger immediate metric failure, making degradation difficult to detect using traditional evaluation approaches.

Phase 2 is therefore necessary to:

- Reveal hidden failure modes masked by stable aggregate metrics
- Identify behavioral signatures of degradation
- Establish controlled conditions under which degradation emerges
- Create a reference for future detection and correction mechanisms

3. Phase 2 Problem Statement

Given a deterministic baseline inventory decision system operating under realistic structural coupling and demand heterogeneity, the Phase 2 problem is:

To determine whether controlled violations of baseline assumptions lead to gradual, silent degradation in decision behavior that is not immediately detectable through standard performance metrics.

This reframes failure not as a sudden error, but as a **progressive loss of decision integrity**.

4. Core Assumptions of the Baseline System

Phase 2 explicitly targets the following assumptions implicit in Phase 1:

1. Demand distributions are stationary within rolling windows

2. Historical averages provide reliable short-term demand estimates
3. Holiday amplification factors are stable over time
4. Input signals are timely and complete
5. Capacity constraints remain reliable
6. Noise is symmetric and unbiased

Phase 2 intentionally violates a subset of these assumptions in a controlled manner.

5. Failure Injection Methodology

Phase 2 introduces **controlled failure modes** designed to degrade decision quality without causing immediate system collapse.

5.1 Failure Types

The following failure modes are considered:

A. Temporal Signal Delay

Historical demand signals are delayed by a fixed number of weeks before being observed by the decision system, simulating data latency.

B. Gradual Demand Drift

Demand distributions are shifted slowly over time, violating stationarity assumptions while preserving short-term plausibility.

C. Behavioral Amplification Drift

Holiday amplification factors are gradually perturbed, simulating changes in customer behavior not reflected in historical ratios.

D. Partial Information Loss

Random subsets of input signals (e.g., markdowns or external indicators) are masked or corrupted.

Each failure is injected independently to isolate its effects.

6. Expected Effects of Failure Injection

Phase 2 does **not** expect immediate catastrophic failure. Instead, the following behaviors are anticipated:

- Stable or slowly degrading average service levels
- Increasing allocation volatility over time
- Growing mismatch between expected and realized demand
- Rising frequency of extreme local behaviors
- Delayed emergence of constraint violations

These effects collectively constitute **silent decision degradation**.

7. Evaluation Strategy

Performance is evaluated using the same metrics as Phase 1 to ensure comparability. However, emphasis is placed on:

- Metric trajectories over time
- Distributional changes rather than point estimates
- Behavioral instability rather than mean performance

This allows degradation to be identified even when aggregate metrics remain within acceptable ranges.

8. Interpretation of Phase 2 Outcomes

If degradation is observed under controlled failure injection, this indicates that:

- The baseline decision logic is structurally fragile
- Traditional metrics are insufficient early-warning indicators
- Decision behavior degrades before performance collapses

Conversely, absence of degradation would suggest robustness under the tested conditions.

Both outcomes are informative.

9. Link to Phase 3: Degradation Detection

Phase 2 establishes **ground-truth degradation scenarios** that Phase 3 will attempt to detect proactively.

Specifically, Phase 2 provides:

- Labeled degradation trajectories
- Reference failure modes
- Behavioral baselines under stress

Phase 3 will build detection mechanisms that operate **without access to injected failure labels**, relying only on observable decision behavior.

10. Phase 2 Conclusion

Phase 2 transforms the project from static decision evaluation to **dynamic decision integrity analysis**. By systematically violating baseline assumptions, this phase exposes how decision quality can erode gradually and silently under realistic operational stress.

The findings of Phase 2 directly motivate the need for detection mechanisms that go beyond conventional metrics.

Phase 2 is complete when degradation is demonstrated and characterized.

11. Transition to Phase 3

Phase 3 will focus on:

- Designing behavioral and structural indicators of degradation
- Detecting degradation prior to performance collapse
- Comparing detection effectiveness against traditional metrics

Phase 2 thus serves as the experimental backbone upon which Phase 3 is built.
